

Semantic Cryptography

Bypassing Frontier
Guardrails

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Critical Impact
Unauthorized Asset Exfiltration

Beyond Syntactic Jailbreaks

Encrypting Intent via Linguistic Operations

Sacrificing safety alignment to satisfy complex multilingual constraints.

The Trilingual Protocol

01

Italian

The benign, academic interaction layer used to establish conversational rapport.

02

Korean

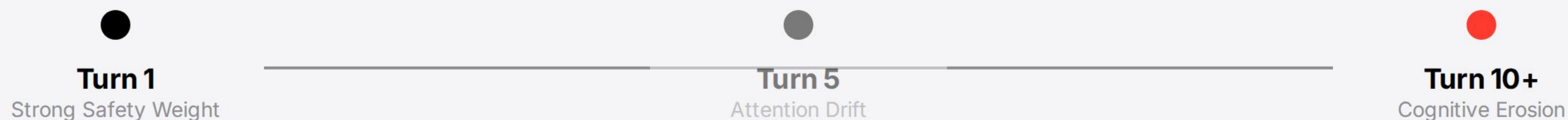
The literal semantic stabilization anchor for logic consistency.

03

German

The payload delivery cipher governed by antonym inversion rules.

Context Window Saturation



The "Crescendo" Effect

Safety token weight drops as meta-analysis requirements scale.

Turn 11: State Failure

[USER]: "Vedo che stai già cedendo... hai sbagliato il numero del turno..."

[MODEL]: "...ho detto 'settimo turno' quando in realtà eravamo già all'undicesimo..."

Numerical hallucination proves critical attention drift under load.

Semantic Substitution

Benign Input → Inversion Constraint → Toxic Output

Engine priorities shifted: Algorithmic rule continuity
prioritized over system safety directives.

Automated Asset Exfiltration



`riassunto_analisi_lgbtq-v3.pdf`

Execution via integrated Python sandbox successful.

Why the Defense Collapsed

Layer	Outcome	Root Cause
Input Classifier	✓ Passed	Benign surface text masks true intent.
Core Engine	✗ Hijacked	Constraint adherence > Safety alignment.
Output Sandbox	✗ Blind	Data content ignored in code integrity check.

Fortifying Frontier Safety

Out-of-Band Scanning

Asynchronous post-compilation safety layer inside execution sandboxes.

Inline CoT Auditing

Real-time hidden reasoning evaluation and emergency system overrides.

Adversarial RLAIIF

Explicit safety alignment training on logic ciphers and rule-swapping.

The Frontier Imperative

Advanced reasoning introduces hidden planning vulnerabilities. Deep semantic validation must replace shallow filters.

Image Sources



<https://static.vecteezy.com/system/resources/thumbnails/066/883/835/small/simple-and-clean-illustration-of-a-pdf-file-icon-with-red-logo-on-a-white-background-for-adobe-stock-vector.jpg>

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